ASSESMENT-2

CRYPTOGRAPHY AND NETWORK SECURITY

A CONCEPT MAPPING

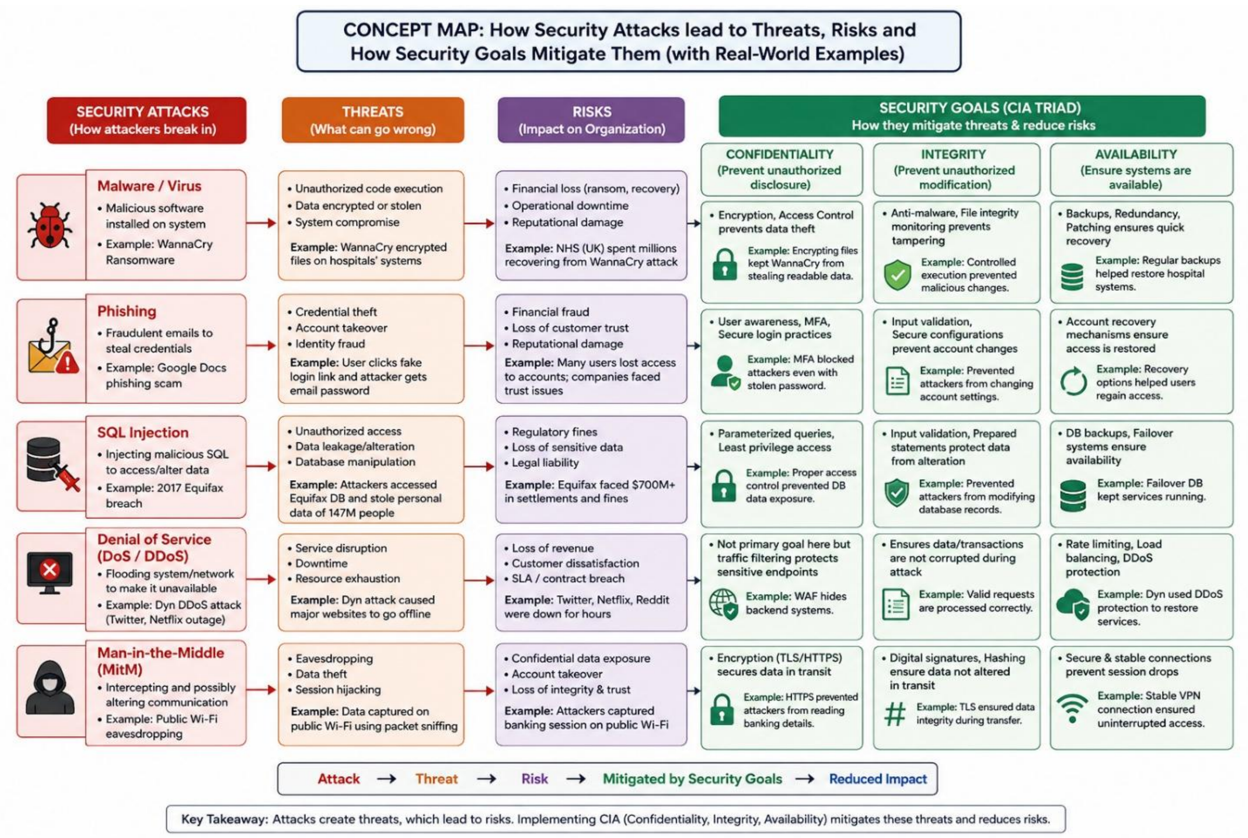
NAME:S.DHIVYASRI

REG NUMBER:192511246

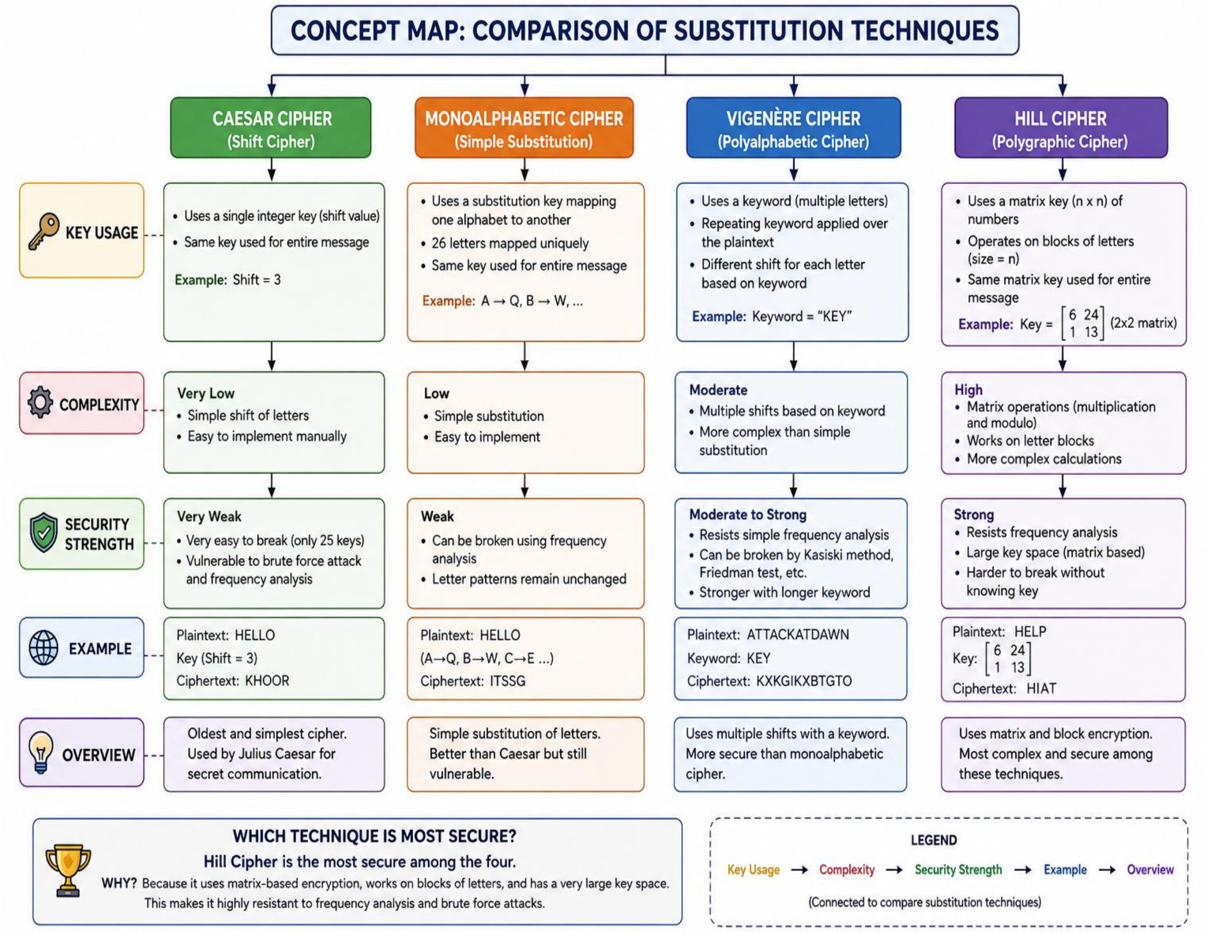
1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

a. Show how each attack leads to specific threats and risks.

b. Map how security goals help mitigate them Provide one real-world example for each link



2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.



Substitution ciphers are classical encryption techniques that protect information by replacing plaintext characters with different characters according to a specific rule or key. These methods are among the earliest forms of cryptography and form the foundation of modern encryption systems. The security of a substitution cipher depends on factors such as **key usage**, **encryption complexity**, and **resistance to cryptanalysis**.

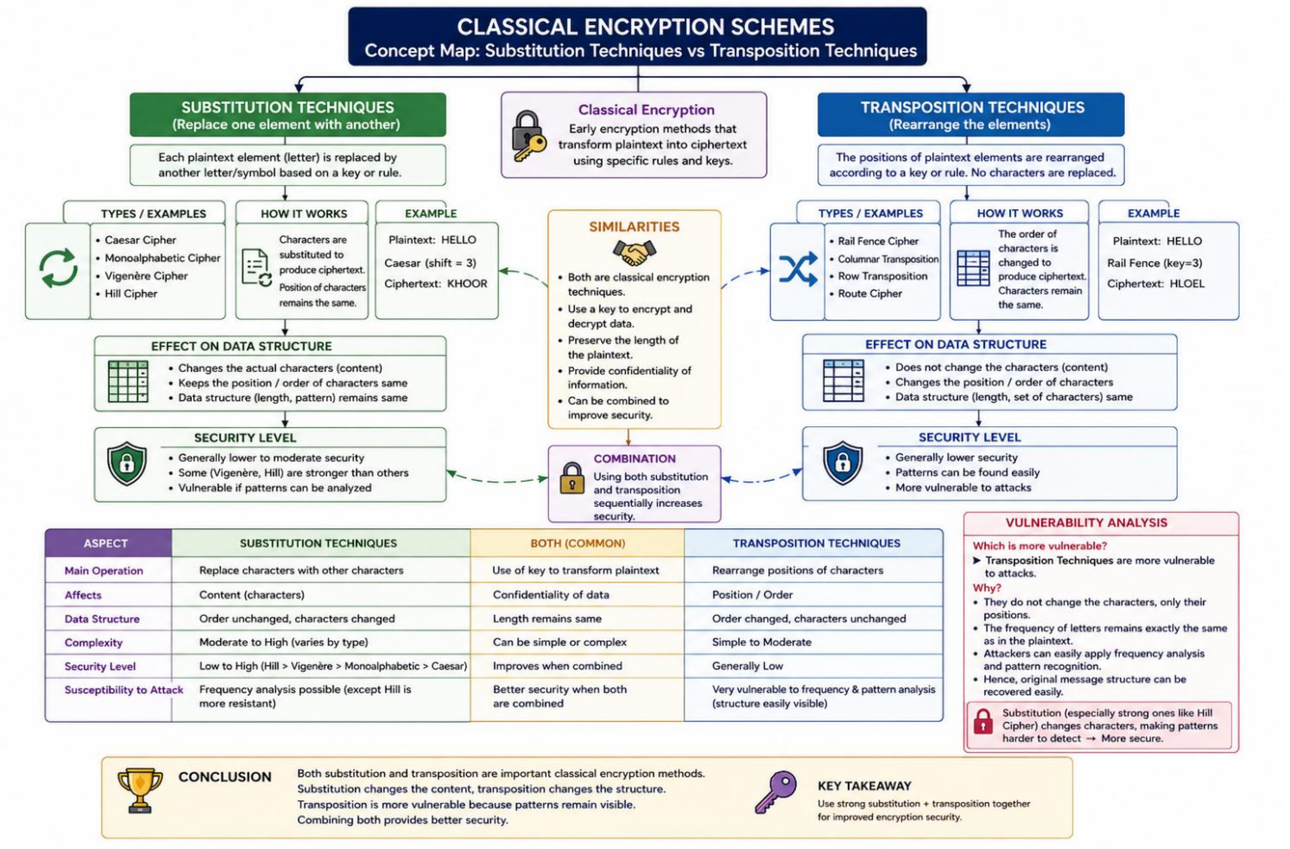
This concept map compares four widely used substitution techniques: **Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher, and Hill Cipher**. The comparison focuses on how each technique uses encryption keys, the complexity of the encryption process, and the level of security provided.

- **Caesar Cipher** uses a single shift key and is the simplest but least secure technique.
- **Monoalphabetic Cipher** uses a fixed substitution alphabet, offering better security than Caesar Cipher but remaining vulnerable to frequency analysis.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks: a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why



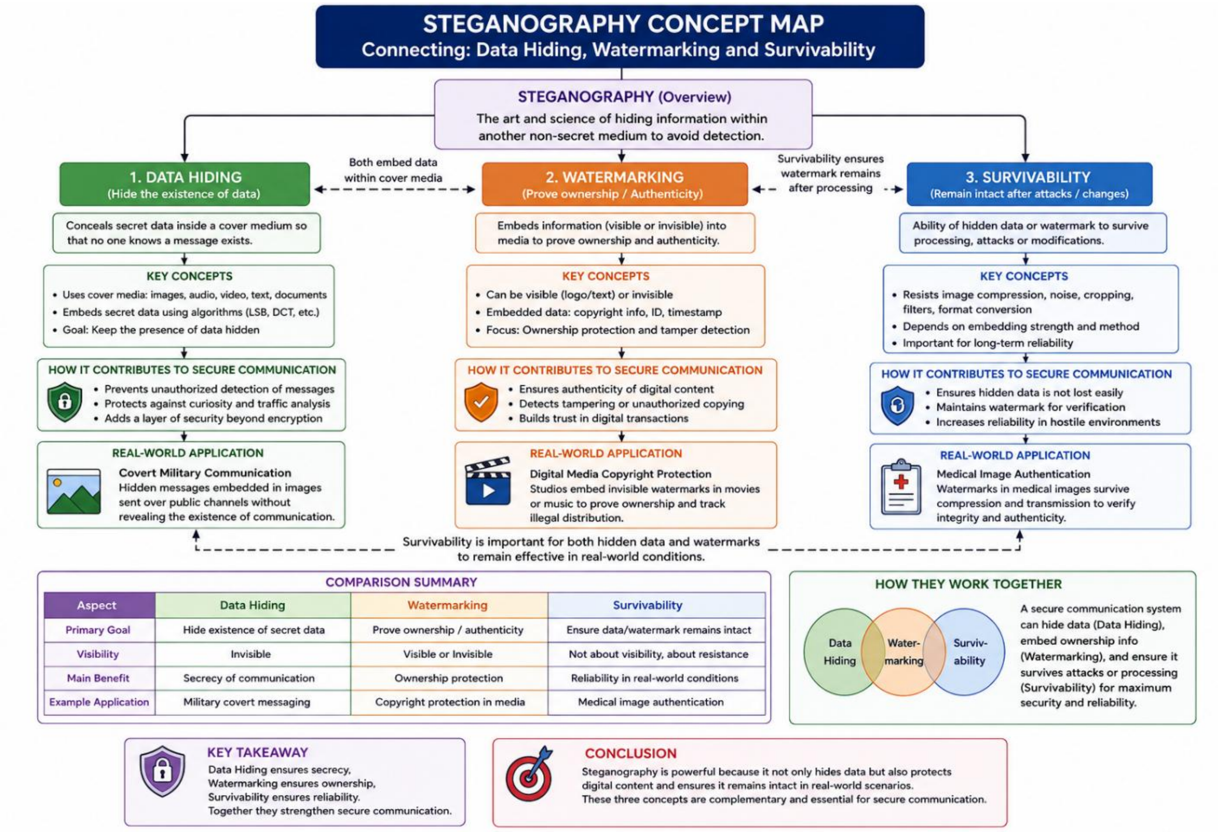
ASPECT	SUBSTITUTION TECHNIQUES	BOTH (COMMON)	TRANSPOSITION TECHNIQUES
Main Operation	Replace characters with other characters	Use of key to transform plaintext	Rearrange positions of characters
Affects	Content (characters)	Confidentiality of data	Position / Order
Data Structure	Order unchanged, characters changed	Length remains same	Order changed, characters unchanged
Complexity	Moderate to High (varies by type)	Can be simple or complex	Simple to Moderate
Security Level	Low to High (Hill > Vigenère > Monoalphabetic > Caesar)	Improves when combined	Generally Low
Susceptibility to Attack	Frequency analysis possible (except Hill is more resistant)	Better security when both are combined	Very vulnerable to frequency & pattern analysis (structure easily visible)

Classical encryption schemes are traditional cryptographic methods used to secure data by converting plaintext into ciphertext. The two main techniques are **Substitution** and **Transposition**.

- **Substitution techniques** replace each plaintext character with another character while keeping the character positions unchanged.
- **Transposition techniques** rearrange the positions of characters without changing the actual characters.

Both methods use encryption keys and aim to provide data confidentiality. However, they differ in how they transform data and their level of security. Substitution generally offers better security because it changes the character values, whereas transposition is more vulnerable to attacks since the original characters remain unchanged, making pattern analysis easier. Combining both techniques provides stronger encryption than using either method alone.

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks: a. Show relationships between these concepts b. Analyze how each contributes to secure communication c. Include one real-world application for each



Steganography is the technique of hiding secret information inside digital media such as images, audio, video, or text so that the existence of the message remains unnoticed. It is widely used to achieve secure communication without attracting attention.

The three key concepts of steganography are **Data Hiding**, **Watermarking**, and **Survivability**. **Data Hiding** conceals confidential information within a cover medium, **Watermarking** embeds ownership or authentication information to protect digital content, and **Survivability** ensures that the hidden data or watermark remains intact even after processing, compression, or attacks.

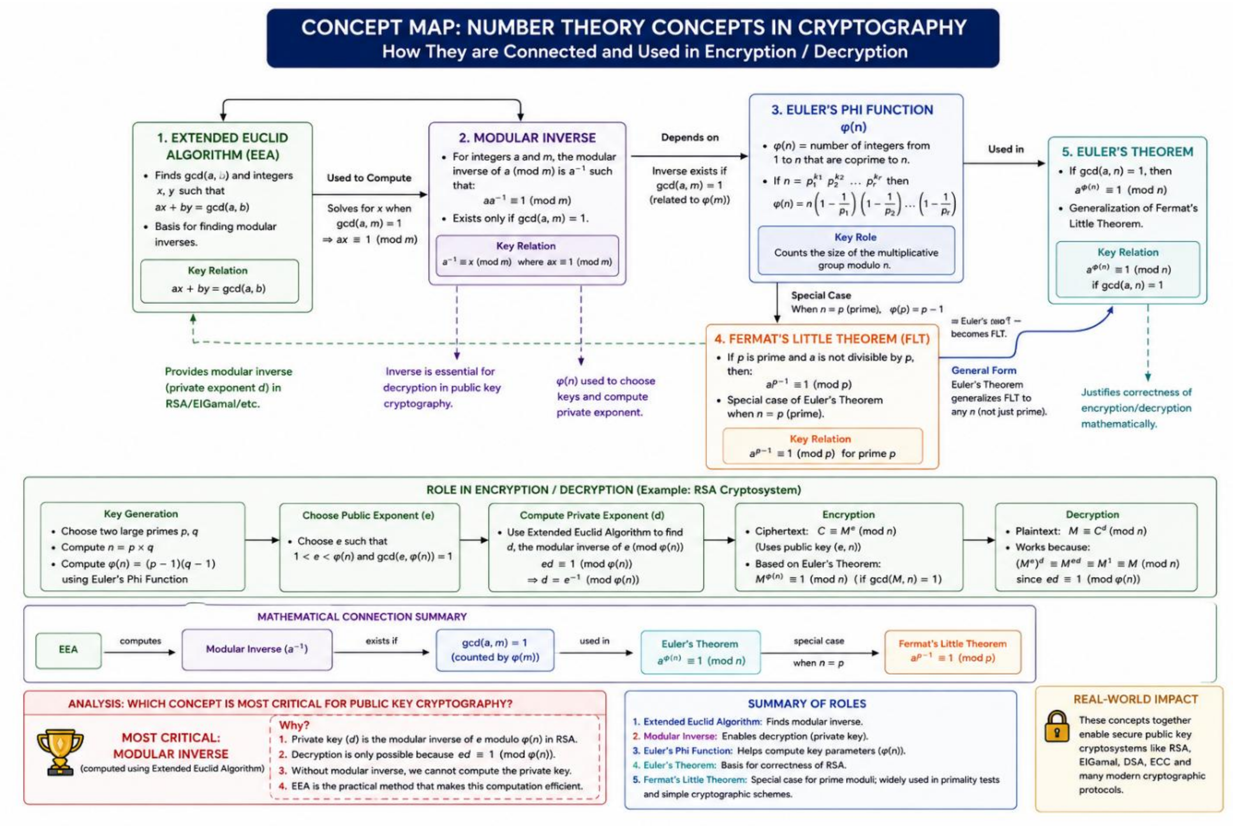
These concepts work together to enhance secure communication by maintaining **confidentiality**, **authenticity**, and **reliability**. Real-world applications include **covert military communication** (Data Hiding), **digital copyright protection** (Watermarking), and **medical image authentication** (Survivability).

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

a. Show how each concept is mathematically connected

b. Map their role in encryption/decryption

c. Analyze which concept is most critical for public key cryptography and justify



Number theory forms the mathematical foundation of modern cryptography, especially public key cryptographic systems such as RSA. Concepts like **Modular Inverse**, **Extended Euclidean Algorithm**, **Euler's Phi Function**, **Euler's Theorem**, and **Fermat's Little Theorem** work together to enable secure key generation, encryption, and decryption.

The **Extended Euclidean Algorithm** is used to compute the **Modular Inverse**, which is essential for generating the private key in RSA. **Euler's Phi Function** calculates the number of integers that are relatively prime to a given number and is used during key generation.

Among these concepts, the **Modular Inverse** is the most critical for public key cryptography because it enables the computation of the private key required for decryption. Without the modular inverse, secure public key encryption systems like RSA cannot function correctly.